

Chinese Activity Constructions: Light Verb Syntax and Adnominalisation:

Abstract:

In Chinese Activity constructions, the Activity nominal (Actor NOMINALIZER Activity) can appear either as the subject or the object of the Activity verb, in the latter case the subject usually consists of the Actor (Actor V Actor NOMINALIZER Activity). While the latter occurs in all Chinese dialects, the former is only found in northern dialects that generalise d-forms of the nominalizer, as southern dialects that use g-forms of the nominalizer prohibit the Activity nominal in subject position. In line with Huang's (2008) influential analysis in which subject Activity nominals are raised from object position, Tang (2008, 2009, 2011) proposes that *ge* blocks verb movement from the embedded clause in the Activity nominal to the matrix verb position, which is supposed to derive the ungrammaticality of *ge* in Activity nominals in subject position. Based on the technical difficulties of Huang and Tang's analyses, this paper proposes a new formal analysis in which Actor and Activity are merged in two separate positions in accordance with thematic principles (Ramchand (2008)), and the microvariation between *de* and *ge* is argued to be due to different nominalization strategies in which the Activity nominal contains either a nominal or clausal complement. The microvariation between *de* and *ge* can hence be derived from different selectional properties, and empirical support is offered from the history of Chinese as *de* and *ge* have different etymologies.

Keywords: Chinese, Mandarin, Cantonese, Activity, Nominalization

Introduction:

Mandarin Chinese has a nominal element *de* (的) known as the adnominalizer or structural particle (結構助詞) which is commonly used to express possession and modification in the noun phrase (Paul (2005, 2012, 2019)). It has been argued that *de* lexicalizes different functional heads in the nominal domain where possession (POSS) and modification (Mod) are hierarchically arranged along with quantificational and indexical categories such as demonstratives, numerals, quantifiers and classifiers (Aoun and Li (2003:140), Del Gobbo (2003:48), Cheng and Sybesma (2015:1520)):

1) Possessive Demonstrative Number Quantifier Classifier Modifier Noun
Mandarin Chinese regularly employs adnominalizer *de* to construct nominal phrases in Activity constructions where the subject of the Activity verb is the Actor and the object is a nominal phrase headed by *de* containing the Actor and the Activity in a pseudo-possessive construction (Actor *de* Activity < Possessor *de* Possessum):

- 2) 他 當 他 的 老師
 ta dang ta de laoshi
 he do he DE teacher
 'He is being a teacher (literally 'he is playing his role as a teacher').'
 (Huang (2008:226), Tang (2011:149))

As the object (e.g. 他的老師 *ta de laoshi* in 2)) denotes abstract activity (here 'him being a teacher') rather than possession ('his teacher'), the head noun (老師 *laoshi* in 2)) does not denote a specific person or entity but an abstract activity (Huang (2008:225), Hu (2016:435)). Furthermore, it is generally possible to use the nominal phrase headed by *de* as the subject of the same Activity verbs, as shown below:

- 3) 他 的 老師 當 得 好
 ta de laoshi dang de hao
 he DE teacher do COMP good
 'He is being a good teacher.' (Huang (2008:225))

While the nominal phrase headed by *de* can occur in either subject (3) or object position (2) of Activity verbs, southern dialects of Chinese that use g-forms of the adnominalizer do not have the nominal phrase denoting Activity occur in the subject position (4b), even though the same nominal phrase can occur in the object position (4a), which points to an important difference between dialectal forms of the Chinese adnominaliser (*de/ge*) (Tang (2008, 2009, 2010, 2011)):

- 4a) 佢 做 佢 嘅 老師
keoi zou keoi ge lousi
he do he GE teacher
‘He is being a teacher.’
- 4b) *佢 嘅 老師 做 得 好
keoi ge lousi zou dak hou
he GE teacher be COMP good
‘He is being a good teacher.’ (Cantonese) (Tang (2008:75, 2009: 241-242, 2011:150))

These discrepancies between adnominalizers *de* and *ge* point to a genuine difference between northern and southern dialects of Chinese, since Activity phrases nominalised by g-forms in southern dialects generally do not occur in the subject position of Activity constructions (5d-f), even though they are widely attested in the object position (5a-c), as confirmed in Chinese dialectal surveys (Tang (2009)):

- 5a) 渠 當 渠 個 老師 (Shandong dialect (Xiang))
keoi dang keoi ge laoshi
he do he GE teacher
- 5b) 伊 當 伊 個 老師 (Taiwanese Min)
yi dang yi ge laoshi
he do he GE teacher
- 5c) 渠 當 渠 個 先生 (Wenzhou dialect (Southern Wu))
keoi dang keoi ge xiansheng
he do he GE teacher
‘He is being a teacher.’ (adapted from Tang (2009:242))
- 5d) *渠 個 老師 當 得 好 (Shandong dialect (Xiang))
keoi ge laoshi dang de hao
he GE teacher do COMP good
- 5e) *伊 個 老師 當 甲 好 (Taiwanese Min)
yi ge laoshi dang ga hao
he GE teacher do COMP good
- 5f) *渠 個 先生 當 好 險 (Wenzhou dialect (Southern Wu))
keoi ge xiansheng dang hao xian
he GE teacher do good PART
‘He is being a good teacher.’ (Tang (2009:241))¹

In sum, while it is acceptable in all Chinese dialects to use adnominalizers *de* and *ge* to nominalise Activity in the object position of Activity verbs (2, 4a, 5a-c), *de* but not *ge* can also head the Activity phrase in the subject position of the same Activity verbs (3, 4b, 5d-f). Adnominalizer *ge* seems more marked than *de*, as the former can only nominalise the Activity phrase in the object position of Activity constructions (4a-b, 5a-f) while the latter can do so in both the subject and object positions (2-3). The two structural minimal pairs of interest here are the subject and object asymmetry in the use of *ge* as the nominalizer of Activity phrases (4a-b, 5a-f) and the contrast between *de* and *ge* as the nominal head of

¹ There are alternative structures in these dialects where subject Activity phrases can be headed not by the adnominaliser (*ge*) but by functors derived from lexical nouns that denote the instrument with which the Activity is carried out e.g. Cantonese:

- 1) 佢 手 籃球 打 得 好
keoi sau laamkau da dak hou
he hand basketball play COMP good
‘He plays basketball well.’
- 2) 佢 腳 波 踢 得 好
keoi geot bo tek dak hou
he foot ball kick COMP good
‘He plays football well.’ (Tang (2009:242))

Activity phrases in subject position (3, 4b), both of which have implications for the Chinese nominal domain.

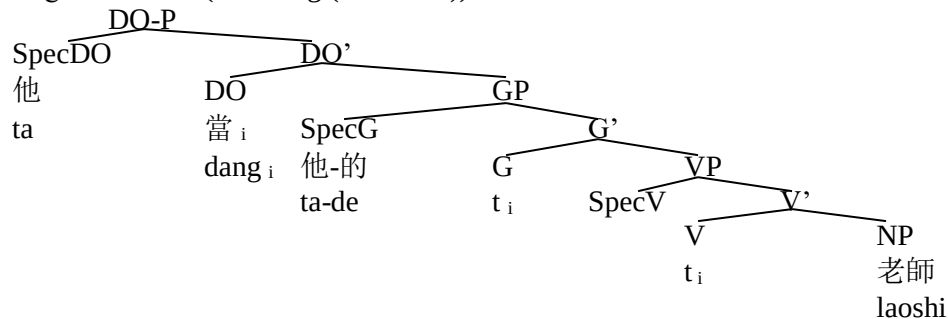
Tang (2008, 2009, 2011) adopts Huang's (2008) analysis of Chinese Activity constructions and proposes a nominalization parameter which seeks to account for the microvariation between *de* and *ge*. Based on the problems in their analyses, this paper proposes a new formal analysis of Chinese Activity constructions that aims to provide an adequate account for the subject and object asymmetry in the dialectal forms of the Chinese adnominalizer (*de/ge*), namely the fact that while both *de* and *ge* can occur in the object position of Activity verbs (2, 4a), only *de* (3) and not *ge* (4b) can occur in the subject position of such verbs, which lies at the heart of the microparametric variation in Chinese dialects. There are four sections in this paper: section 1 critically reviews Huang (2008) and Tang (2008, 2009, 2010, 2011) and shows some of their weaknesses, namely Huang's stipulation that the Activity nominal moves from the object position to the subject position, which is not only unnecessary but is also at odds with traditional Projection Principles that assume a one-to-one correspondence between thematic arguments and A(argument)-positions (Hu (2016:437), cf Chomsky (1981:36), Stowell (1981:112)), and Tang's nominalization parameter, which does not adequately account for the prohibition of *ge* as the head of Activity nominals in subject position. Section 2 proposes modifications to Huang and Tang's analyses by eliminating the *ad-hoc* unjustified movement of the Activity phrase. Rather, it seems more plausible that Actor and Activity are merged in two distinct A-positions in line with traditional thematic hierarchies (Baker (1985, 1988)). Since Actor is the external argument in Activity constructions (Hale and Keyser (1993:77-84, 2002:88-103)), Actor and Activity can be argued to be merged in the specifier positions of Ramchand's (2008) InitiateP and ProcessP respectively, which conforms to the event structure of Activity where SpecInitiateP and SpecProcessP hold the agent and the theme respectively (Ramchand (2008:52ff)). The canonical base structure of Chinese Activity constructions, therefore, consists of Actor and Activity as two independent arguments, which provides a new derivational model (Initiate + Process) for examining the microparametric variation between adnominalizers *de* and *ge*. Section 3 reconsiders the microvariation between *de* and *ge* in light of the new event structure where Actor and Activity are merged in SpecInitiateP and SpecProcessP respectively, and the subject and object asymmetry of *de/ge* in heading the Activity phrase can be remodelled thus: both *de* and *ge* can head the Activity phrase (SpecProcessP) if there is an Actor argument (SpecInitiateP) projected above it, but in the absence of such Actor argument (SpecInitiateP), only *de* and not *ge* can head the Activity phrase (SpecProcessP). In sum, *ge* can head the Activity phrase (SpecProcessP) iff the Actor argument is projected (SpecInitiateP), which indicates a fascinating and inextricable link between the Actor argument (SpecInitiateP) and the Activity phrase headed by *ge* (SpecProcessP) and forms the crux of the dialectal microvariation in Chinese Activity constructions. As the Actor argument seems to impose selectional restrictions on the nominalising head of the Activity phrase (*de/*ge*), it may be argued that Actor is base-generated in the Activity phrase (SpecProcessP) and raised to its surface matrix A-position (SpecInitiateP) via some kind of A(argument)-copying as seen in the coreference between the subject of Activity constructions and the external argument of the Activity phrase (他_i 當他_i 的老師 *ta_i dang ta_i de laoshi* (2) / 佢_i 做佢_i 嘅老師 *keoi_i zou keoi_i ge lousi* 'he is being a teacher' (4a)). In the absence of the matrix Actor argument, on the other hand, there is a fundamental difference in the Activity phrase which can only be nominalised by *de* and not by *ge* (他的老師當得好 *ta de laoshi dang de hao* (3) /* 佢嘅老師做得好 *keoi ge lousi zou dak hou* 'he is being a good teacher' (4b)), and this can be correlated with the etymological origins of *de* and *ge* as *de* is widely taken to derive from phrase-final nominaliser *di* (底) in nominal linking contexts (Feng (1990), Jiang (1999), cf den Dikken and Singhapreecha (2004)) whereas *ge* is derived from the general classifier *ge* (個/固) which may be incompatible with nouns denoting abstract Activity as classifiers typically have count, individualising and delimiting properties that naturally select concrete and countable nouns (Bisang and Li (2012), Zhang (2013)). The microvariation between *de* and *ge* in Activity phrases can hence be reconsidered in light of a biclausal structure where the matrix Actor argument (SpecInitiateP) is extracted from the Activity phrase (SpecProcessP) that should contain an embedded clause in a case of clausal nominalization. In the absence of the Actor argument, the Activity phrase can be argued to contain not a

clausal but a nominal complement, which is subject to the selectional differences between *de* and *ge*, the latter of which may be incompatible with lexical nouns denoting abstract Activity due to its de-classifier origins. Section 4 concludes by examining this biclausal structure of Chinese Activity constructions (Actor and Activity merged in SpecInitiateP and SpecProcessP respectively), which conforms to Lin's (2001) Light Verb syntax that argues for a rich inventory of Light Verbs in Chinese serving as argument-introducing heads (cf Huang (1994, 1997)). Furthermore, the correlation between the matrix Actor argument and the internal structure of the nominalised Activity phrase indicates that while the Activity argument may be thematically constant in being merged in the same A-position (SpecProcessP), it can have a different internal structure in containing either a nominal or clausal complement, which differs from traditional distinctions between result and process nominalisations (Grimshaw (1990)).

Section 1: Chinese Activity Constructions (Huang (2008)) and Nominalization Parameter (Tang (2008, 2009)):

Following influential accounts by Hale and Keyser (1993, 2002)), Huang (2008:226-232) argues that Light Verb DO selects a nominalised gerund phrase (GP) headed by adnominalizer *de*, which selects a verbal complement denoting Activity, and when the verb in GP undergoes head movement to DO via the head of GP (G), this derives the canonical word order where the Actor and Activity phrase co-occur as the subject and object of the Activity verb respectively:

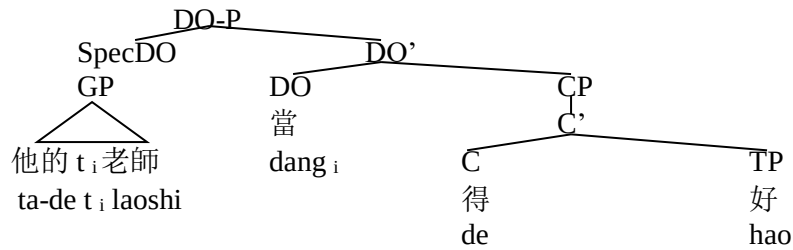
- 6) 他 當 他 的 老師
 ta dang ta de laoshi
 he do he DE teacher
 'He is being a teacher.' (cf Huang (2008:230))



Moreover, Huang (2008:230-232) argues that when the matrix Actor argument (他 *ta* in 6)) is deleted, it is possible for the verbless Activity GP (他的 *t_i* 老師 *ta de t_i laoshi* in 6)) to undergo remnant movement to the specifier of DO (SpecDO-P), which derives the subject and object alternation of the Activity phrase as it now occurs in subject position (2-3). This also leaves room for the Activity verb to select a different complement such as the adverbial phrase 得好 *dehao* 'good, well' (7) (Huang (2008:230-232)):²

- 7) 他 的 老師 當 得 好
 ta de laoshi dang de hao
 he DE teacher do COMP good
 'He is being a good teacher.' (Huang (2008:231))

² Chinese regularly uses non-thematic complements to express adverbial notions such as duration, frequency and extent (Li (1990:7-10)) which, in accordance with Projection Principles such as Baker's UTAH, are merged below all thematic arguments of the verb (Lin (2001)).

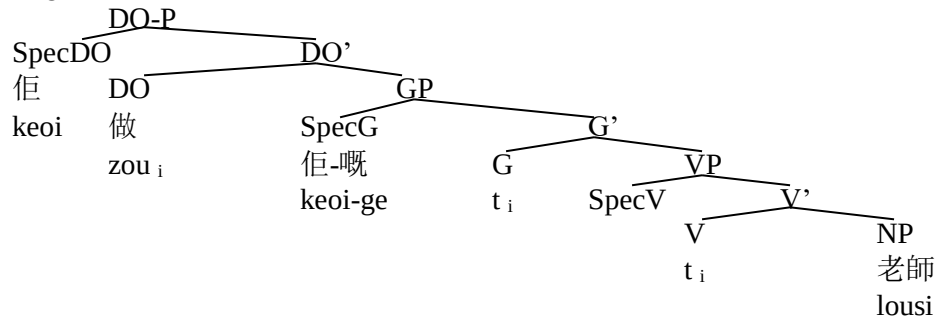


Adopting Huang's derivational model, Tang (2008, 2009) proposes a nominalization parameter where the markedness of Cantonese *ge* is accounted for in terms of selectional properties, since it is argued that while Mandarin *de* can select clausal complements (8a) (Si (2002)), Cantonese *ge* cannot (8b). This is taken to support a nominalization parameter where *de* but not *ge* permits verb head movement from GP to DO via G and hence account for why *de* but not *ge* can nominalize Activity phrases in subject position (8c-d) (Tang (2008:76-77, 2009:243-244)):

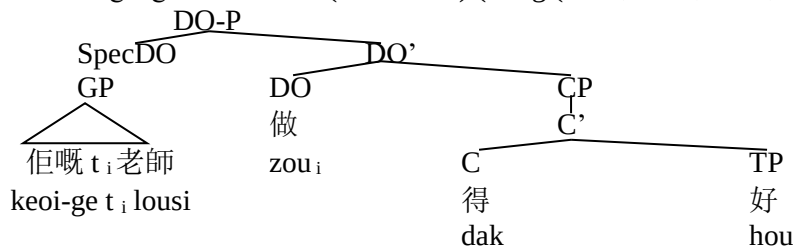
- 8a) 張三 的 曾 在 美國 留學
 Zhangsan de ceng zai meiguo liuxue
 Zhangsan DE once at America study.abroad
 'Zhangsan's studying in the US in the past' (Si (2002:35))

- 8b) *張三 嘅 曾經 响 美國 留學
 Zeongsaam ge cangging heong meigwok lauhok
 Zeongsaam GE once at America study.abroad
 'Zeongsaam's studying in the US in the past' (Tang (2008:77))

- 8c) 佢 做 佢 嘅 老師
 keoi zou keoi ge lousi
 he be he GE teacher
 'He is being a teacher.'



- 8d) *佢 嘅 老師 做 得 好
 keoi ge lousi zou dak hou
 he GE teacher be COMP good
 'He is being a good teacher.' (Cantonese) (Tang (2008, 2009, 2010, 2011))



There are numerous problems in Huang and Tang's analyses that require scrutiny and modification. First of all, while Huang seeks to unify Chinese Activity constructions with Activity constructions in other languages such as English and Japanese denominal constructions by postulating a Light Verb projection

DO that introduces the Actor (SpecDO-P) and triggers head movement, the postulation of a nominal Activity phrase (GP) as the complement to DO seems to be a categorial mistake, as Light Verbs are merged above verbal projections and should select verbal complements (Chomsky (1993), Hale and Keyser (1993, 2002), Cinque (1999)). Secondly, the categorial and syntactic status of adnominaliser *de* (6) and by extension *ge* (8c) is unclear, since Huang's (2008:230) argues that *de* is cliticised onto the Actor argument in SpecGP (6), which permits head movement of the embedded verb to DO via G (當 *dang*_i ... t_i in 6), 做 *zou*_i ... t_i in 8c)). However, this clitic analysis of *de* is counterintuitive to the conventional analysis of *de/ge* as nominal(ising) heads (1) (Tang (2006), Li (2008)), in which case *de/ge* (G) should block head movement of the embedded verb to DO by principles of locality and minimality (Rizzi (1990)). Thirdly, the subject and object alternation of the Activity phrase (7, 8d) is not independently motivated as no explanation is given for the deletion of the Actor argument in SpecDO-P and the subsequent remnant movement of the Activity phrase, which seems to be an *ad-hoc* circular argument for accounting for the empirical data.³ Finally, and most crucially, it is unclear how Tang's nominalization parameter accounts for the minimal contrast between *de* (7) and *ge* (8d) in the Activity phrase in subject position, since even if *de* and *ge* were both cliticised onto the Actor argument in SpecGP and permitted V-to-G-to-DO movement, it is unclear how and why *ge* but not *de* would block such head movement without positing extra and unnecessary stipulations. Furthermore, the raised Activity phrase can only be nominalised by *de* and not by *ge* (他的老師 *ta de laoshi*/*佢嘅老師 *keoi ge lousi* in 10f)), which Tang (2008 :76-77, 2009 :243-244) seeks to account for with his nominalization parameter that states that only *de* and not *ge* permits the embedded verb to move to DO via G (t_i in 10e)). In addition to the problem of how *de* and *ge* might block V-to-G-to-DO head movement, it is unclear how Tang's nominalization parameter differentiates between the subject and object positions of the Activity phrase, since the derivation predicts that remnant movement of the Activity phrase is possible if it is headed by *de* but impossible if it is headed by *ge*, which is left unexplained. The dialectal microvariation between *de* and *ge* in the subject-object asymmetry in the nominalization of Chinese Activity phrases is a real empirical phenomenon which remains to be explained, and the formal and empirical properties of Chinese adnominalizers (*de/ge*) and Activity constructions are presently explored.

Section 2: Chinese Activity constructions: Actor + Activity:

As shown above, all Chinese dialects have Activity constructions in which Actor and Activity occur in the subject and object positions of the Activity verb respectively (2, 4a), which is repeated here (9a-b):

- 9a) 他 當 他 的 老師
 ta dang ta de laoshi
 he do he DE teacher (Mandarin)
- 9b) 佢 做 佢 嘅 老師
 keoi zou keoi ge lousi
 he do he GE teacher (Cantonese)
 'He is being a teacher.' (Huang (2008:226), Tang (2011:149))

Huang (2008:231-232)) supposes the deletion of the Actor argument in SpecDO-P and remnant raising of the Activity phrase to the same A-position (see footnote 3), which is shown here again (cf 6-8):

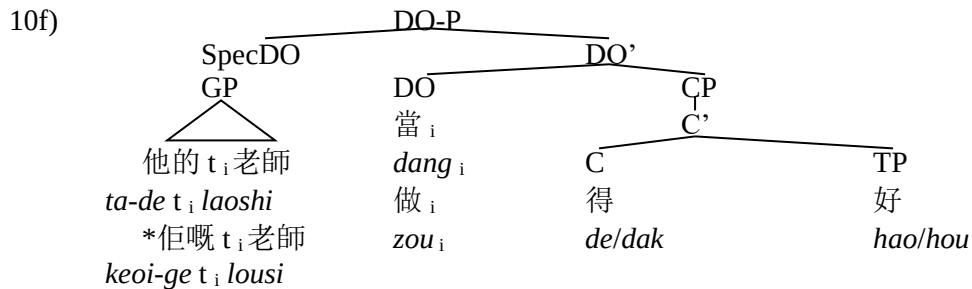
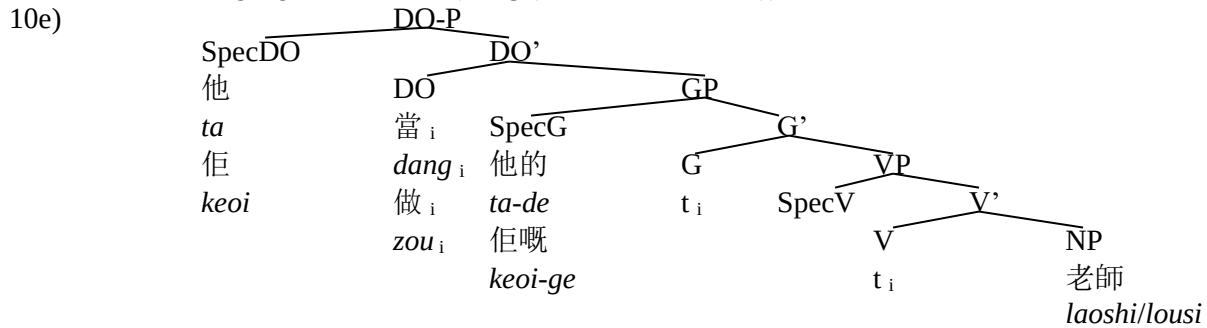
- 10a) 他 當 他 的 老師
 ta dang ta de laoshi
 he do he DE teacher (Mandarin)

³ Professor Jim Wood (p.c.) has suggested to me that this alternation may be on a par with passivisation where the subject argument is suppressed and the object argument is moved to the subject position for Case-assignment (Burzio (1986)). However, while the subject/object alternation of the Activity phrase is schematically similar to the Case-driven raising of the object in passivisation, there is not an independent principle such as Case to derive it (cf Chomsky (1981)), which leaves it as yet unexplained.

10b) 佢 做 佢 嘅 老師
 keoi zou keoi ge lousi
 he do he GE teacher (Cantonese)
 ‘He is being a teacher.’ (Huang (2008:225-226), Tang (2011))

10c) 他 的 老師 當 得 好
 ta de laoshi dang de hao
 he DE teacher do COMP good (Mandarin)

10d) *佢 嘅 老師 做 得 好
 keoi ge lousi zou dak hou
 he ge teacher do COMP good (Cantonese)
 ‘He is being a good teacher.’ (Tang (2008:75, 2011:150))



As shown in 10f), when the Actor argument (他 *ta*/佢 *keoi* in 10e)) is deleted and the verbless Activity GP undergoes remnant movement (他的 t_i 老師 *ta de t_i laoshi*/*佢嘅 t_i 老師 *keoi ge t_i lousi* in 10f)) to SpecDO-P, it is possible for Light Verb DO to select a different complement such as adverbial complement phrase 得好 *de hao/dak hou* ‘good, well’ (10f) (see footnote 1). This is problematic, since Actor and Activity may co-occur with non-thematic complements such as adverbial clause 得好 *dehao* ‘good, well’ (11a-b), which renders the deletion of the Actor and the remnant movement of the Activity phrase unnecessary as they may be base-generated in two different A-positions that dominate non-thematic complements such as the adverbial clause (see footnote 2):

11a) 他 當 他 的 老師 當 得 好
 ta dang ta de laoshi dang de hao
 he do he DE teacher do COMP good (Mandarin)

11b) 佢 做 佢 嘅 老師 做 得 好
 keoi zou keoi ge lousi zou dak hou
 he do he GE teacher do COMP good (Cantonese)
 ‘He is being a good teacher.’⁴

⁴ It has been remarked to me by other native speakers that examples like 11a-b) sound a little stilted and should be marked by ?. Nonetheless, it remains the case that it is possible to express Actor and Activity together with an adverbial complement, as seen in the following examples where the Activity phrase is simplified by omitting the possessor (= Actor) of the phrase:

[illegible]

13)

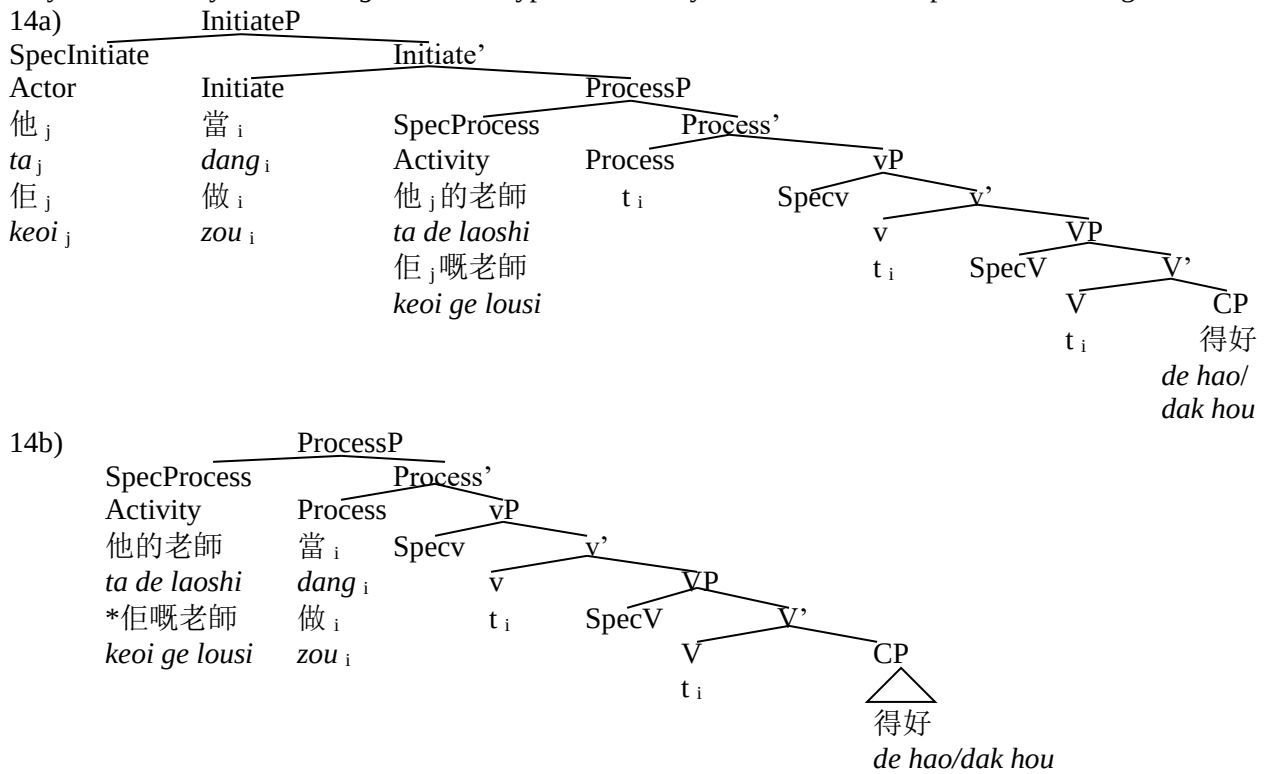
	ProcessP	
SpecProcess	Process'	
Activity	Process	vP
他的老師	當 _i	Specv v'
<i>ta de laoshi</i>	<i>dang_i</i>	v VP
*佢嘅老師	做 _i	t _i SpecV V' CP
<i>keoi ge lousi</i>	<i>zou_i</i>	V t _i 得好 <i>de hao/dak hou</i>

i)	他	當	老師	當	得	好
	ta	dang	laoshi	dang	de	hao
	he	do	teacher	do	COMP	good
ii)	佢	做	老師	做	得	好
	keoi	zou	lousi	zou	dak	hou
	he	do	teacher	do	COMP	good
	'He is being a good teacher.' (Mei (1978))					

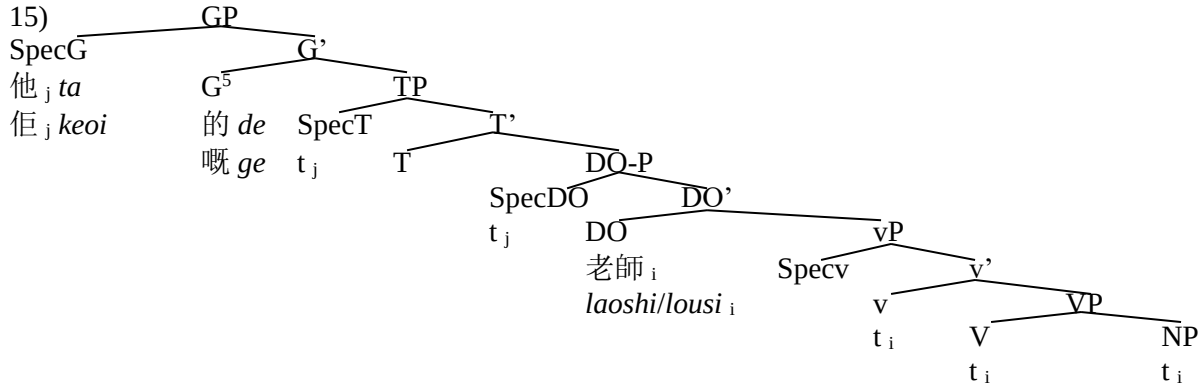
between the matrix Actor argument (SpecInitiateP) and the Activity phrase (SpecProcessP), since in the absence of the former the latter can only be headed by *de* and not by *ge* (13). It follows that the projection of the matrix Actor argument (SpecInitiateP) renders possible the nominalization of the Activity phrase (SpecProcessP) by *de* and, more importantly, by *ge* (12), and the internal structure of the Activity phrase seems sensitive not only to the selectional properties of the adnominaliser (*de/*ge*) but also to the projection of the matrix Actor argument, which forms the core of Chinese Activity constructions. The next section provides an explanation for this by considering the internal structure of Chinese Activity phrases.

Section 3: Chinese adnominalisers *de* and *ge* : Activity Phrase:

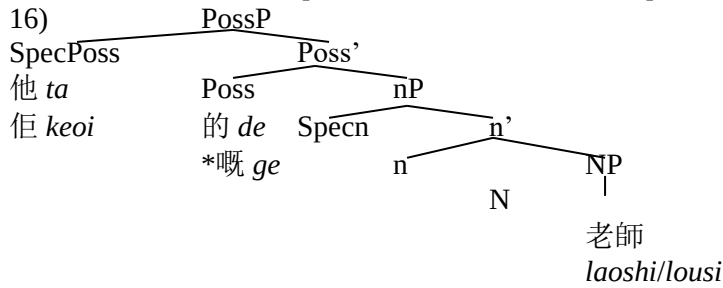
The previous section defended a bi-argumental structure for Chinese Activity constructions where the Activity phrase (SpecProcessP) is dominated by the Actor argument (SpecInitiateP) (12), and this is derived by head movement of the Activity verb (V-to-Process-to-Initiate) that licenses the arguments in the specifier positions of the Light Verb projections. When the matrix Actor argument is not projected (13), the Activity phrase (SpecProcessP) becomes the sole argument in subject position, and here it can only be headed by *de* and not *ge*. The two types of Activity construction are represented here again:



The two minimal pairs of interest are the subject and object asymmetry of the Activity phrase which can only be headed by *ge* (佢嘅老師 *keoi ge lousi*) in object (14a) but not in subject position (14b), and in the subject position the Activity phrase (14b) can only be headed by *de* (他的老師 *ta de laoshi*) and not by *ge* (*佢嘅老師 *keoi ge lousi*). This indicates a different internal structure of the Activity phrase (SpecProcessP) that correlates with the projection of the Actor argument (SpecInitiateP). In line with conflation where the event structure of Activity (DO) triggers head movement that licenses the Actor argument in the specifier of the Light Verb projection (Hale and Keyser (1993)), the presence of the matrix Actor argument (SpecInitiateP) coreferential with the external argument of the Activity phrase (SpecProcessP) suggests that the latter contains an embedded clause in which the former is base-generated in the same functional projection that mirrors the matrix clause. The derivation of the canonical word order of Chinese Activity constructions where both Actor and Activity are expressed is hence biclausal and the Activity phrase here is a case of clausal nominalisation, as shown below:

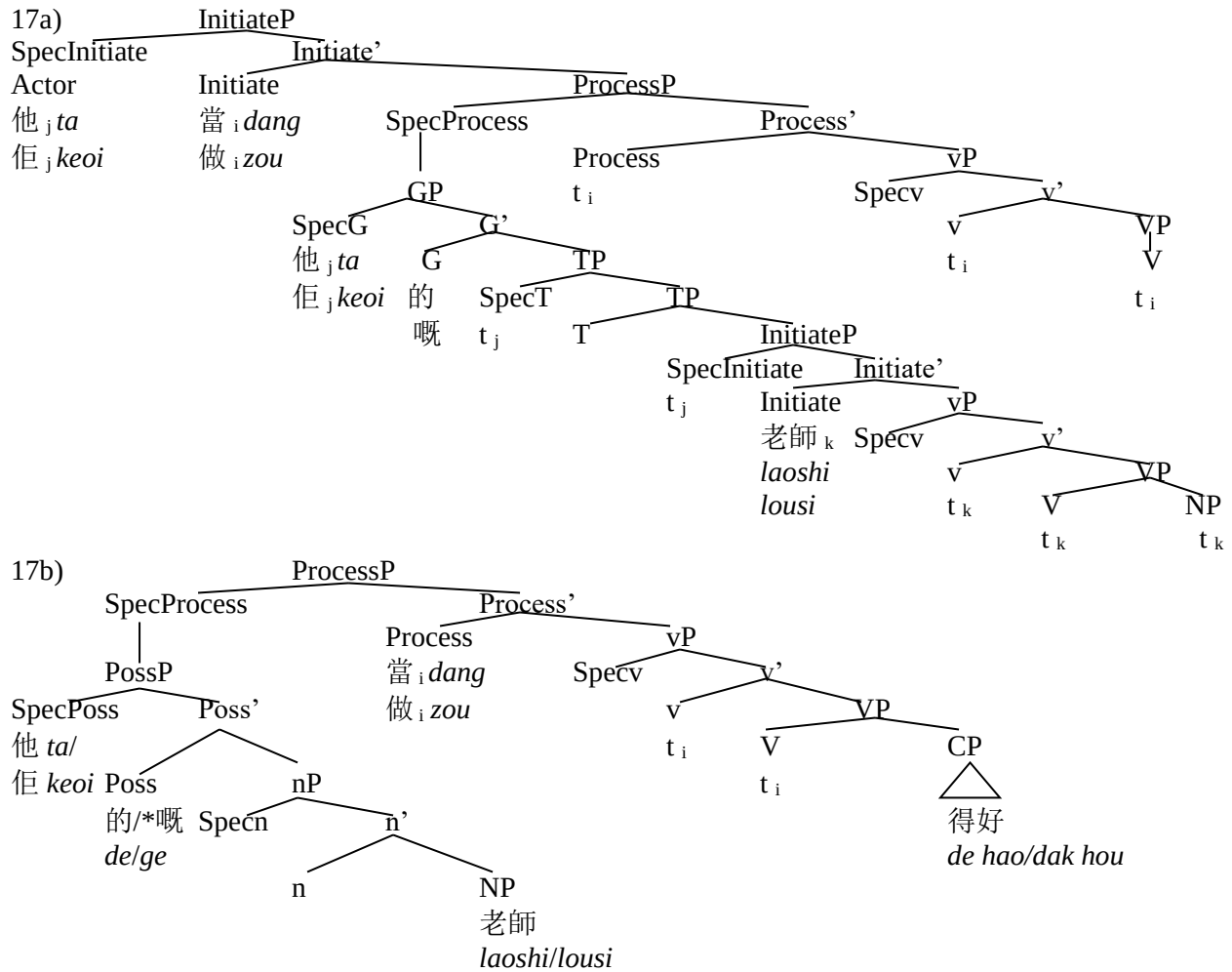


In the absence of the Actor argument, there is no conflation and corresponding head movement and the Activity phrase has a different internal structure as it can only be headed by *de* and not by *ge*. It can be argued that the Activity phrase is of a different type of nominalization as it consists of a more canonical nominal structure with a possessor and a nominal complement denoting Activity, as shown below:



The two types of Activity construction consist of two different types of nominalization, namely clausal nominalization (15) in the case where the matrix Actor argument is projected (14a) and a nominal possessive phrase (16) if the matrix Actor argument is not projected (14b), which correlates with head movement in conflation and indicated by the microvariation between adnominalizers *de* and *ge* in heading the Activity phrase in subject position (13). In sum, while both *de* and *ge* can nominalize a clause by selecting a clausal complement (14a), only *de* and not *ge* can select a nominal complement denoting Activity (14b), which shows that adnominalisers can have selectional restrictions on the lexical noun, even if these do not apply to their clausal complements. The two types of Activity construction can hence be represented as follows:

⁵ Huang's (2008) labelling of the nominal head of clausal nominalizations as G(erund) corresponds to the various D-projections in the nominal domain which also introduce arguments in their specifiers: in 14a) Actor is base-generated in the embedded SpecInitiateP and moved to SpecGP while in 14b) G corresponds to Poss which introduces the possessor to the noun phrase (Longobardi (2001), Longobardi and Silvestri (2013)).



The selectional differences between adnominalisers *de* and *ge* in selecting a noun denoting Activity (17b) can be derived from their etymological and historical origins: adnominalizer *de* is historically derived from phrase-final nominaliser *di* (底) which has been argued to originate from Classical Chinese *zhe* (者) as it is similarly phrase-final (Liu (2008), Aldridge (2009), Yap et al (2011)). The adnominalizing functions of *de* have been shown to derive from appositional constructions where phrase-final *di* is juxtaposed with a coreferential noun phrase which can be reanalysed as a single noun phrase where *di* is reanalysed as a phrase-medial nominal linker anticipating modern Chinese *de*:

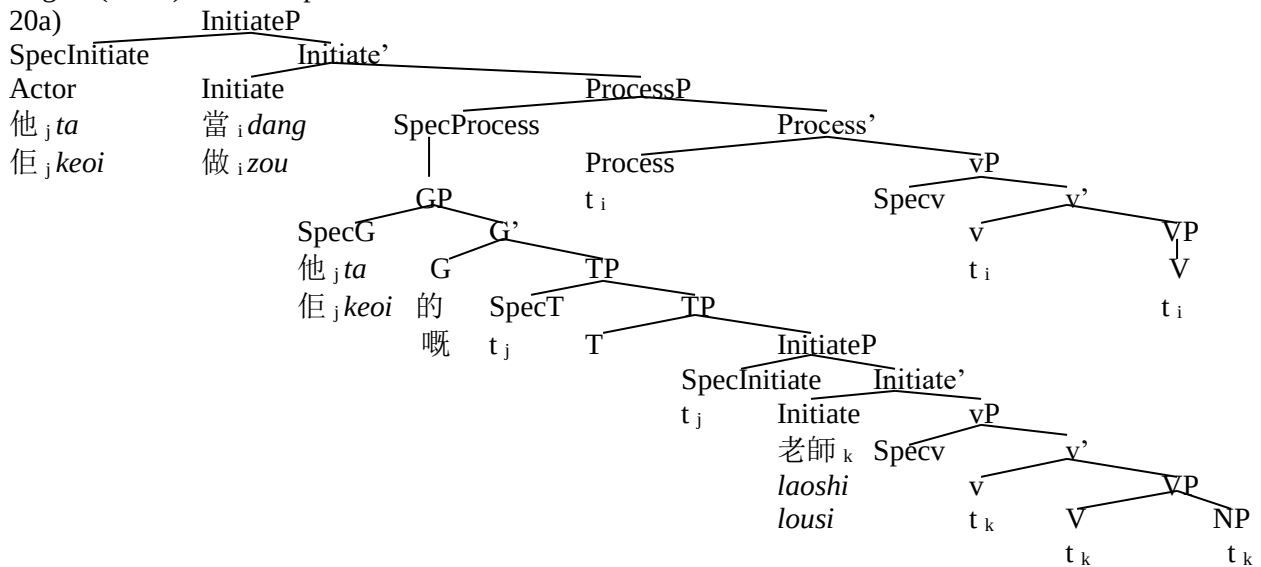
- 18) 昨 來 底 後生
 zuo lai di housheng
 yesterday come DI youth
 ‘the one who came yesterday, the youth’ > ‘the youth who came yesterday’ (祖堂集)

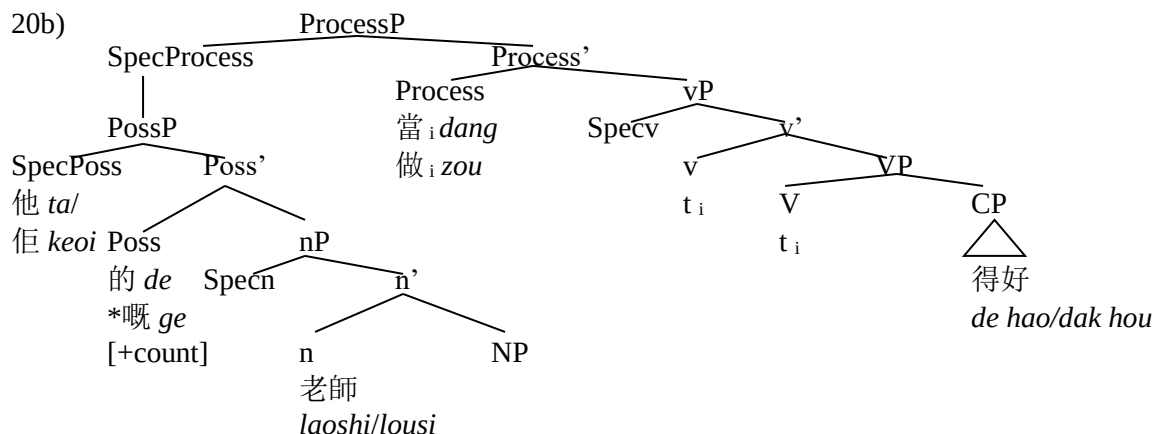
Adnominalizer *ge*, on the other hand, is commonly derived from the general classifier *ge* (個/固) which is typically used for counting and quantification and hence inherently delimits its nominal complement (Bisang and Li (2012), Zhang (2013)). There are hence selectional differences between *de* and *ge*, since while the former is attested with abstract nouns denoting skill/ability (19a-b), the latter regularly selects concrete countable nouns (19c-e) (Cao (1995), Zhao (1999), Shi (2012), Li (2016)):

- 19a) 自己的 鐮 與 槍 舞 得 好
 ziji de jian yu qiang wu de hao
 self DE arms and spear wield COMP good
 ‘He wields his arm and spears well’ (literally ‘his skills in using arms and spear are good’)
 (隋唐演義)

- 19b) 他 的 棋子 雖然 下 得 極 高
 ta de qizi suiran xia de ji gao
 he DE chess though play COMP very good
 'although he plays chess extremely well' (二十年目睹之怪現狀)
- 19c) 好 個 聰明 人
 hao ge congming ren
 good GE clever person
 'A good clever person' (敦煌變文集)
- 19d) 堂堂 好 個 丈夫-兒
 tangtang hao ge zhangfu-er
 indeed good GE man-DIM
 'good men indeed' (敦煌變文集)
- 19e) 後面 個 僧 祇 對 看
 houmian ge seng zhi dui kan
 behind GE monk only face look
 'the monk behind only looks in the face' (景德傳燈錄)

While *ge* regularly selects countable and delimited nominal complements (人 'person' (19c), 丈夫兒 'man' (19d), 僧 'monk' (19e)), *de* selects nouns that can be interpreted as abstract skill and activity (自己 鑊與槍舞得好 'his arms and spear-wielding of arms and spears is good' (18a), 他的棋子下得極高 'his chess-playing skills are very expert' (18b). These selectional and collocational differences seem to have persisted in the dialectal forms of the adnominalizer (*de* and *ge*) which show similar discrepancies in heading Activity phrases in the subject position where the adnominalizer selects nominal complements to which these selectional properties ([+count]) apply, whereas in clausal nominalization such as Activity phrases in the object position, these selectional restrictions seem to be non-applicable as both *de* and *ge* can head Activity phrases (15a, 16a). The semantic markedness of *ge*, which can be traced to its classifier origins (19c-e), can be represented as follows:





Here adnominalizers *de* and *ge* have different selectional properties as *ge* has countable features ([+count]) which restrict its selection of lexical nouns to concrete, countable nouns and prohibit its selection of abstract nouns denoting Activity (e.g. 老師 in 20b)). The next concluding section compares the two types of Chinese Activity constructions and the distribution of *de* and *ge*, which sheds new light on the derivation of Activity constructions and nominalisation.

Section 3: Chinese Activity Constructions: Light Verb Syntax and Nominalization:

This paper analyses the subject and object alternation of Activity phrases in Chinese Activity constructions, especially the subject-object asymmetry between *de* and *ge* in heading the Activity phrase in different positions (introduction). In light of the weaknesses in Huang (2008) and Tang's (2008, 2009, 2011) analysis (section 1), a biclausal and bi-argumental representation is proposed for canonical Chinese Activity constructions where Actor and Activity are overtly expressed as the subject and object of the Activity verb respectively, and these are argued to be two distinct arguments merged in two different A-positions (SpecInitiateP and SpecProcessP respectively) based on thematic principles and event structure (section 2). When Actor and Activity are both present, the former (SpecInitiateP) is argued to be copied and raised from the latter (SpecProcessP), which contains a clausal complement and is hence a case of clausal nominalization (20a). On the other hand, when the Activity phrase (SpecProcessP) is the sole argument of the Activity verb, there is no extraction of the Actor argument and the Activity phrase contains a nominal complement, which is indicated by the selectional differences between the adnominalizers (*de*/**ge*) (20b).

While Chinese lacks derivational morphology, it is possible to trace the category (nominal/clausal) of the complement of the adnominaliser (*de/ge*) in accordance with the selectional restrictions of the adnominalizer, since *ge*, whose featural content ([+count]) can be derived from its classifier origins, is incompatible with nominal complements denoting abstract Activity (section 3). The internal structure of the Activity phrase (clausal nominalization/nominal complement) seems to correlate with the matrix Light Verb projections (Initiate + Process), since the projection of the Actor argument (SpecInitiateP) not only triggers verb head movement (V-to-Process-to-Initiate) in conformity with Hale and Keyser's (1993, 2002) conflation but also entails a biclausal structure where the Activity phrase contains a clausal complement in which the Actor argument is base-generated and extracted from. The categorial ambiguity of the nominalised Activity phrase, as shown by the minimal pairing of the Activity phrase headed by *ge* in the subject and object positions, is dependent not on the Activity phrase, which remains thematically and positionally constant in the specifier of Process, but on the derivation of the Actor argument, which is an interesting innovation to traditional assumptions that nominalisation can be structurally defined in terms of its semantic properties (e.g. Grimshaw's (1990:45ff) result vs event nominalisation). Rather, Chinese Activity constructions show syntactic processes, namely the projection of Actor (SpecInitiateP) and lack thereof, that can affect the internal structure of Activity phrase.

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